



Engineering Electromagnetics

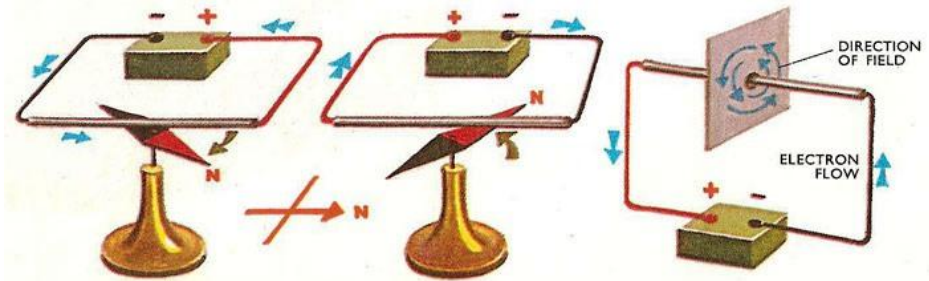
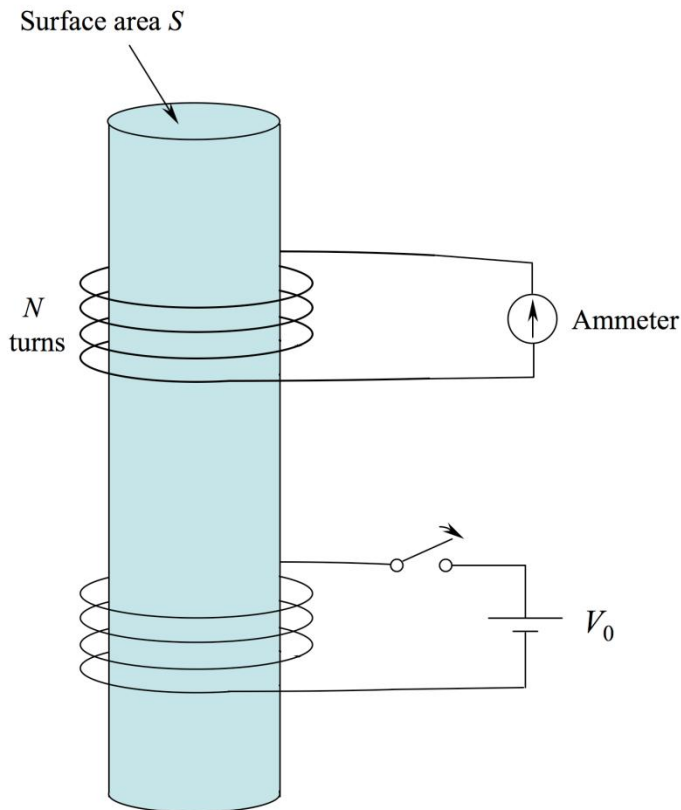
工程电磁场

Youwei Jia 嘉有为

12: Time-Varying Fields 时变场



- Faraday's Experiment

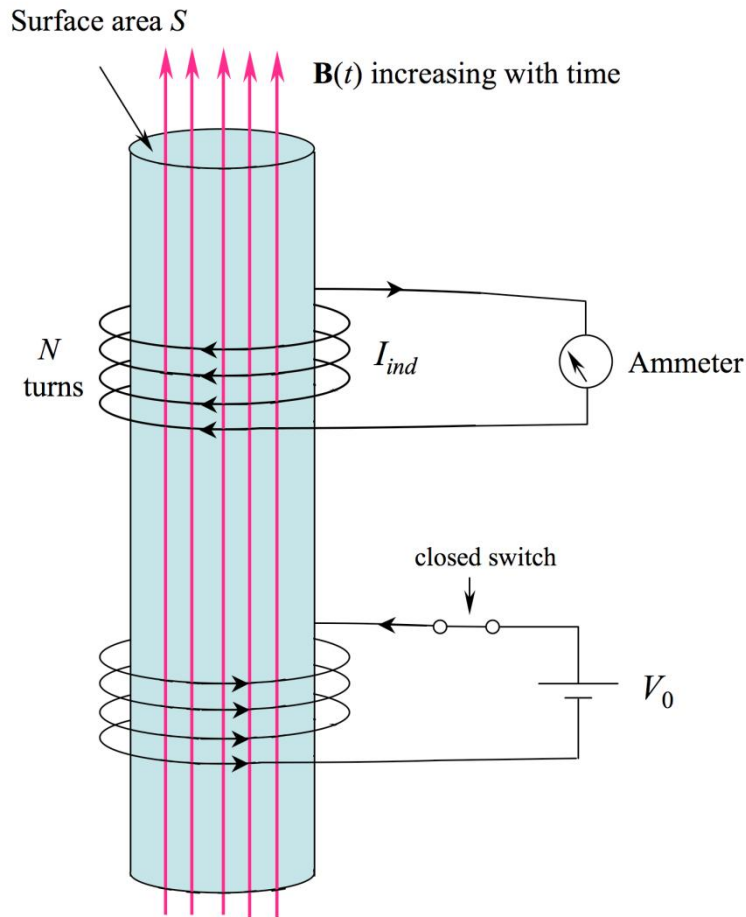


The question addressed is: **If a current can generate a magnetic field, then can a magnetic field generate a current?**

An experiment similar to that conducted to answer that question is shown here. Two sets of windings are placed on **a shared iron core**. In the lower set, a current is generated by closing the switch as shown. In the upper set, any induced current is registered by the **ammeter**.

12: Time-Varying Fields

- Faraday's Experiment- Effect of Increasing $\mathbf{B}$



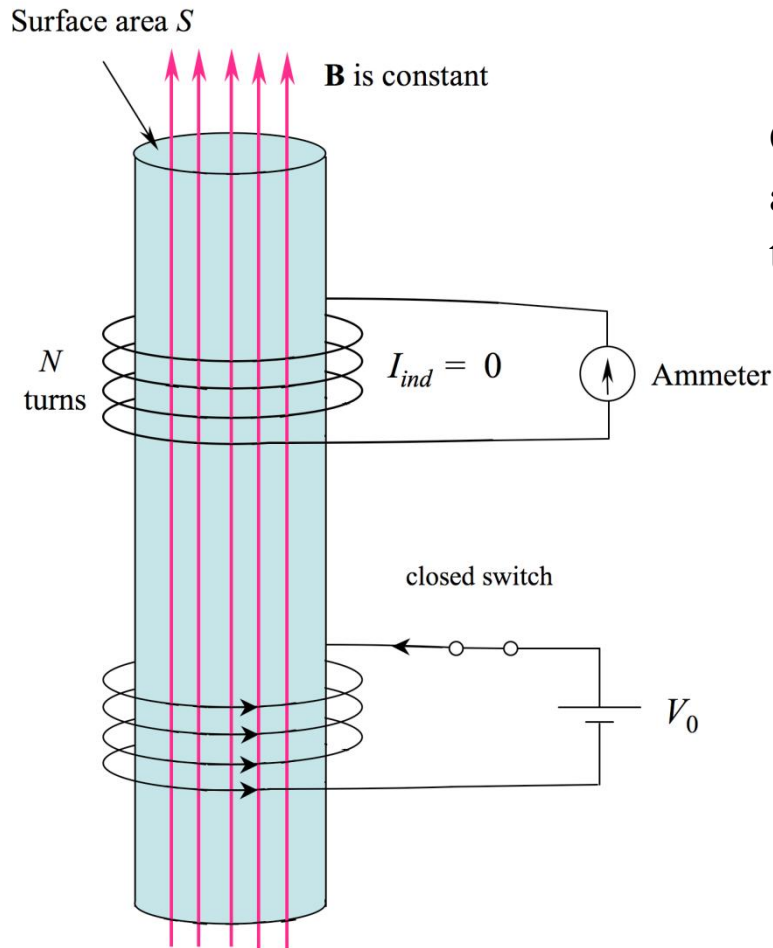
Closing the switch results in current that initially **increases with time in the lower windings**. Magnetic flux density $\mathbf{B}$ is generated, which now links the upper windings; these are found to carry current, I_{ind} . This current exists **as long as $\mathbf{B}$ increases with time**. Note the **direction of the induced current** and the position of the ammeter.

More specifically, **the induced current is proportional to the time rate of change of the magnetic flux** (the surface integral of $\mathbf{B}$ over S).

12: Time-Varying Fields



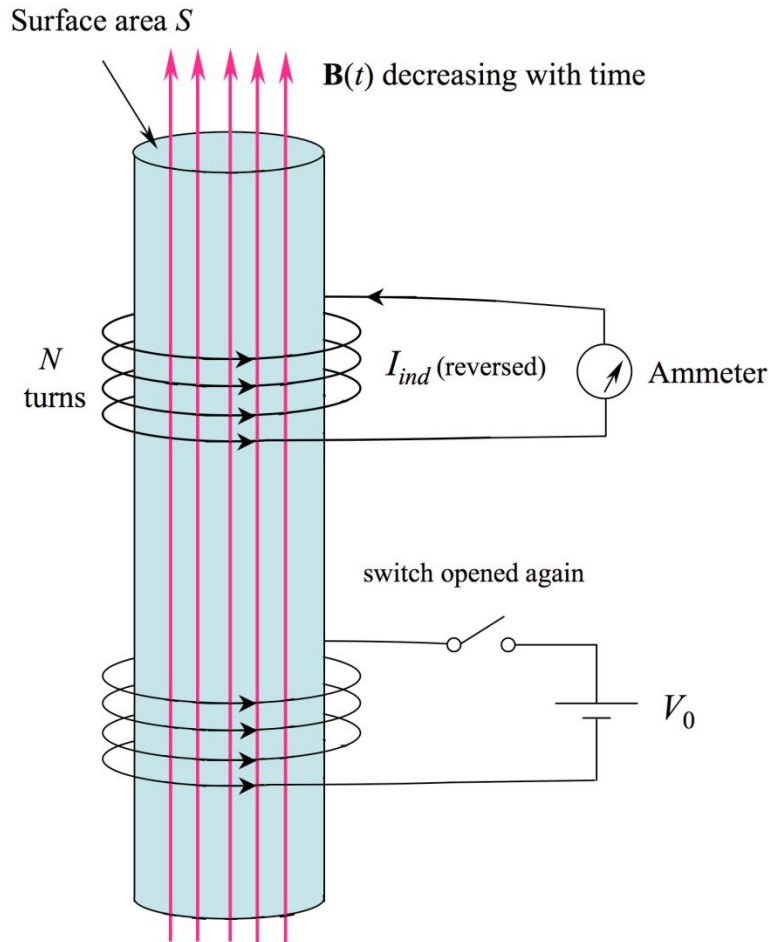
- Faraday's Experiment- Steady State



Once **steady state** is reached, in which $\mathbf{B}$ is at a constant value, the induced current is found to be *zero*.

12: Time-Varying Fields

- Faraday's Experiment- Effect of Decreasing B

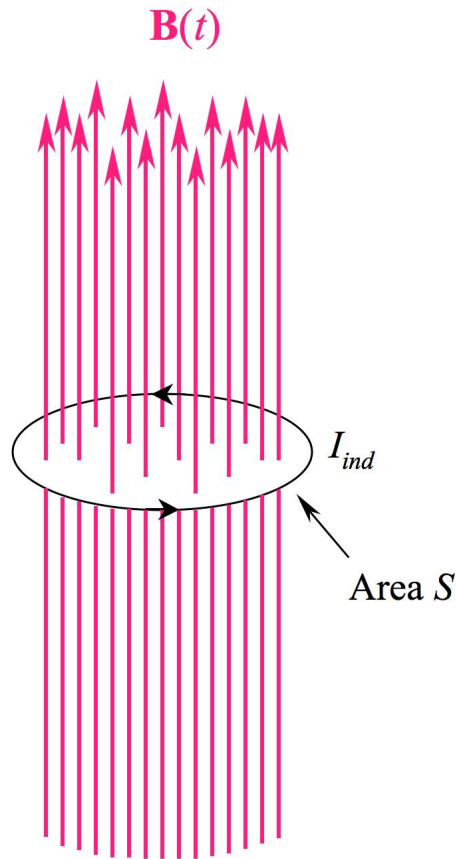


Opening the switch results in current that initially decreases with time in the lower windings. **Magnetic flux density B collapses**, and the upper windings carry current, I_{ind} in the opposite direction from before. This current exists *as long as the magnetic flux decreases with time*.

Once B is reduced completely to zero, the induced current is zero as well.

12: Time-Varying Fields

- A Simpler Demonstration

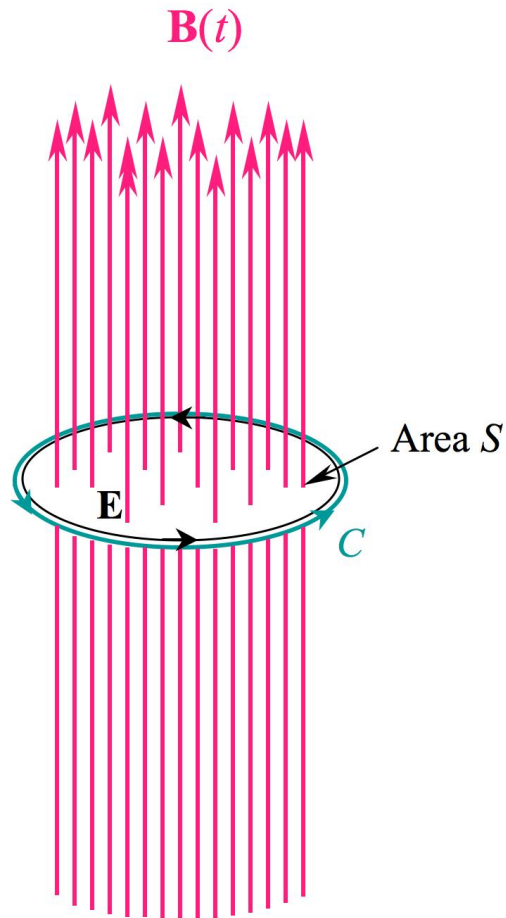


Consider a single turn of wire, through which **an externally-applied magnetic flux is present**. The flux varies with time.

A current, I_{ind} , is generated in the wire loop **as a result of the changing magnetic flux**.

12: Time-Varying Fields

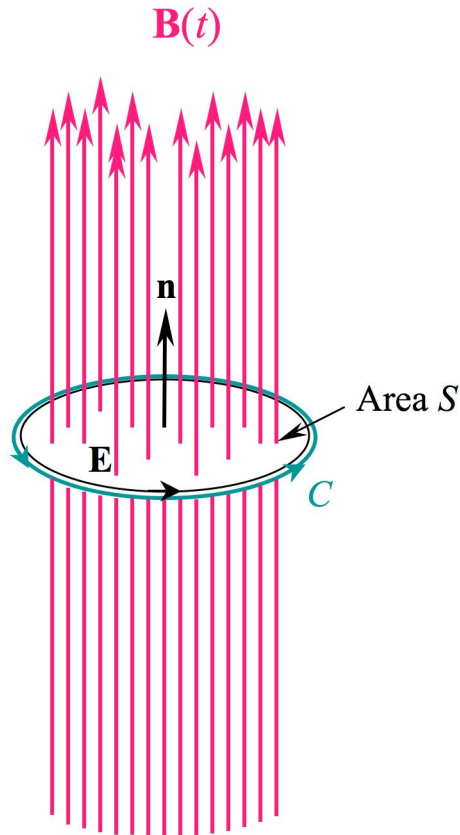
- Replacing Current with **Electric Field**



Current in the wire is caused by electric field, E , that pushes the charge around the wire. In this illustration, the wire is removed, and the electric field remains. An integration contour, C , is also shown, that coincides with E .

12: Time-Varying Fields

- Electromotive Force (emf) 电动势



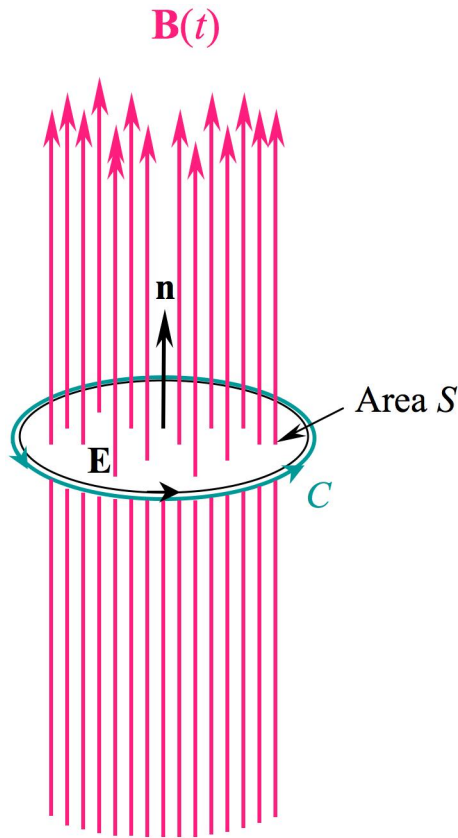
Next, a unit **normal vector to the surface**, $\mathbf{n}$, is shown. The relative directions of $\mathbf{n}$, and the contour $\mathbf{C}$ are defined by the **right-hand convention**: Right hand thumb in direction of $\mathbf{n}$; fingers then curve in the direction of $\mathbf{C}$.

The **electromotive force, or emf**, is defined as the closed path integral of $\mathbf{E}$ about $\mathbf{C}$:

$$\text{emf} = \oint_C \mathbf{E} \cdot d\mathbf{L}$$

12: Time-Varying Fields

- Faraday's Law of Induction 电磁感应定律



Faraday's Law states that the induced emf around a closed path is equal to the time rate of change of the magnetic flux through the area surrounded by the path:

$$\oint_C \mathbf{E} \cdot d\mathbf{L} = -\frac{d\Phi}{dt}$$

where the flux is:

$$\Phi = \int_S \mathbf{B} \cdot d\mathbf{S} = \int_S \mathbf{B} \cdot \mathbf{n} da$$

Note the use of the normal unit vector here, which determines the sign of the flux, and ultimately the emf.

12: Time-Varying Fields

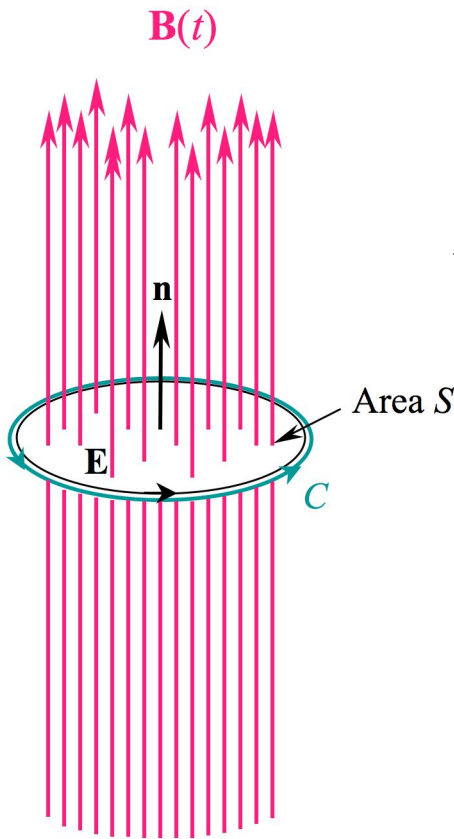
- Faraday's Law in Detail

Using the definitions of emf and the magnetic flux in this situation:

$$\oint_C \mathbf{E} \cdot d\mathbf{L} = -\frac{d\Phi}{dt}$$

becomes:

$$\text{emf} = \oint_C \mathbf{E} \cdot d\mathbf{L} = -\frac{d}{dt} \int_S \mathbf{B} \cdot \mathbf{n} da$$



12: Time-Varying Fields

- Alternative Expression for Faraday's Law
(Another Maxwell Equation)

First, move the time derivative into the interior of the right-hand integral:

$$\Rightarrow \text{emf} = \oint \mathbf{E} \cdot d\mathbf{L} = - \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S}$$

Second, use **Stokes' theorem** to write the line integral of $\mathbf{E}$ as the surface integral of its curl:

$$\Rightarrow \int_S (\nabla \times \mathbf{E}) \cdot d\mathbf{S} = - \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S}$$

Third, simplify by noting that both integrals are taken over the same surface:

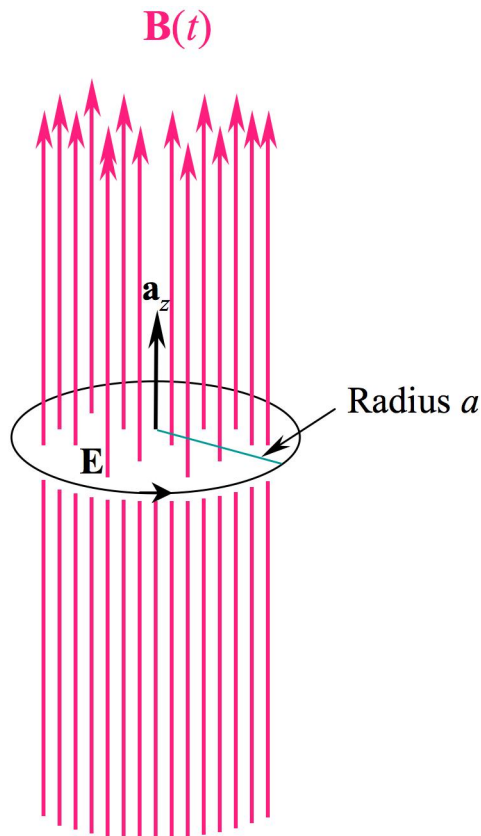
$$\Rightarrow (\nabla \times \mathbf{E}) \cdot d\mathbf{S} = - \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S}$$

Finally, obtain the point form of Faraday's Law:

$$\Rightarrow \boxed{\nabla \times \mathbf{E} = - \frac{\partial \mathbf{B}}{\partial t}}$$

12: Time-Varying Fields

- Example



Start with a uniform (but time-varying) field:

$$\mathbf{B} = B_0 e^{kt} \mathbf{a}_z$$

$$\text{Then } \text{emf} = \oint \mathbf{E} \cdot d\mathbf{L} = 2\pi a E_\phi$$

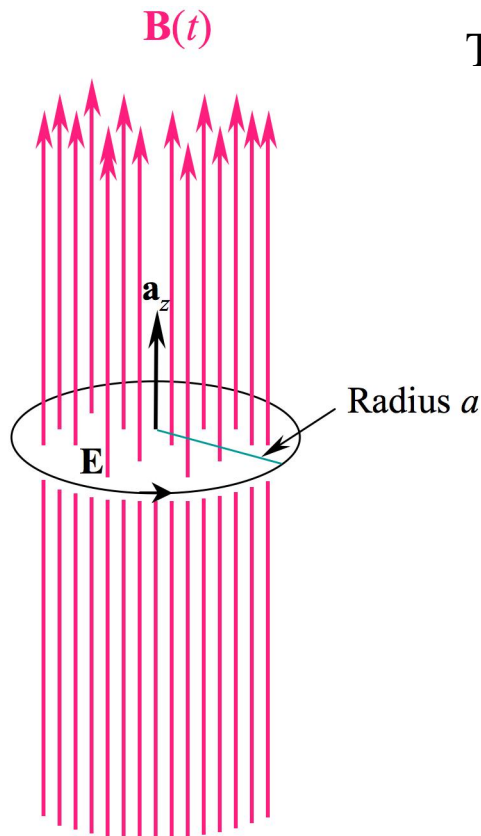
$$= - \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S} = -k B_0 e^{kt} \pi a^2$$

As this calculation will work for any radius, we can replace the fixed radius with a variable radius and write:

$$\mathbf{E} = -\frac{1}{2} k B_0 e^{kt} \rho \mathbf{a}_\phi$$

12: Time-Varying Fields

- Example: Another Way...



This time use: $\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$ where $\mathbf{B} = B_0 e^{kt} \mathbf{a}_z$

$$\text{Therefore: } (\nabla \times \mathbf{E})_z = -k B_0 e^{kt} = \underbrace{\frac{1}{\rho} \frac{\partial(\rho E_\phi)}{\partial \rho}}$$

Surviving z-directed
curl component with
radial variation only

Multiply through by ρ and integrate both sides:

$$-\frac{1}{2} k B_0 e^{kt} \rho^2 = \rho E_\phi$$

Obtain, as before: $\mathbf{E} = -\frac{1}{2} k B_0 e^{kt} \rho \mathbf{a}_\phi$

12: Time-Varying Fields

- Induced emf from a Moving Closed Path

In the case shown here, the flux within a closed path is increasing with time as a result of the **sliding bar**, moving at constant **velocity** v .

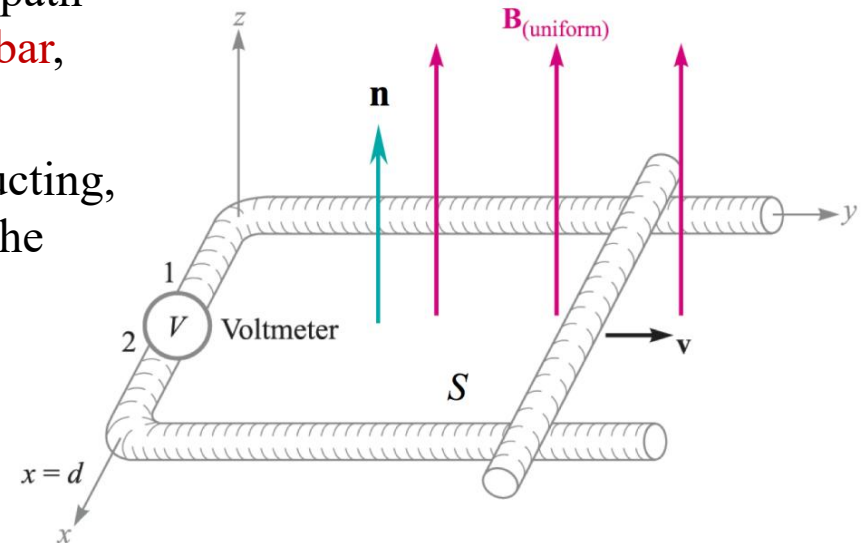
The conducting path is assumed perfectly-conducting, so that the induced emf appears entirely across the voltmeter.

The magnetic flux enclosed by the conducting path is

$$\begin{aligned}\Phi &= \int_S \mathbf{B} \cdot d\mathbf{S} = \int_S \mathbf{B} \cdot \mathbf{n} da \\ &= Byd\end{aligned}$$

where the y dimension varies with time.

Therefore:
$$\text{emf} = -\frac{d\Phi}{dt} = -B\frac{dy}{dt}d = -Bvd$$



12: Time-Varying Fields

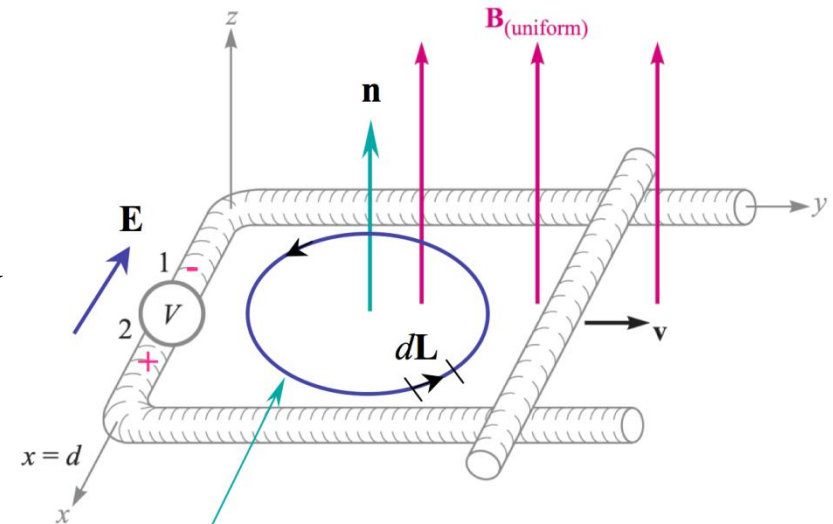
- Finding the Direction of $\mathbf{E}$

..so we have:

$$\text{emf} = -\frac{d\Phi}{dt} = \oint \mathbf{E} \cdot d\mathbf{L} = -Bvd$$

The $\mathbf{E}$ integration path direction is determined by the right hand convention (thumb of right hand in **direction of $\mathbf{n}$** , fingers in direction of closed path). This is the **direction of $d\mathbf{L}$** as shown in the figure.

From the equation, we see that $\mathbf{E} \cdot d\mathbf{L}$ is negative, meaning that $\mathbf{E}$ will point from terminals 2 to 1 as shown. As we have a perfectly-conducting path, $\mathbf{E}$ will exist only at the voltmeter.



Direction of integration path of $\mathbf{E}$ through conductor (by right hand convention)

12: Time-Varying Fields

- Motional emf 动生电动势

The sliding bar contains free charges (electrons) that experience Lorentz forces as they move through the $\mathbf{B}$ field:

Recall: $\mathbf{F} = Q\mathbf{v} \times \mathbf{B}$

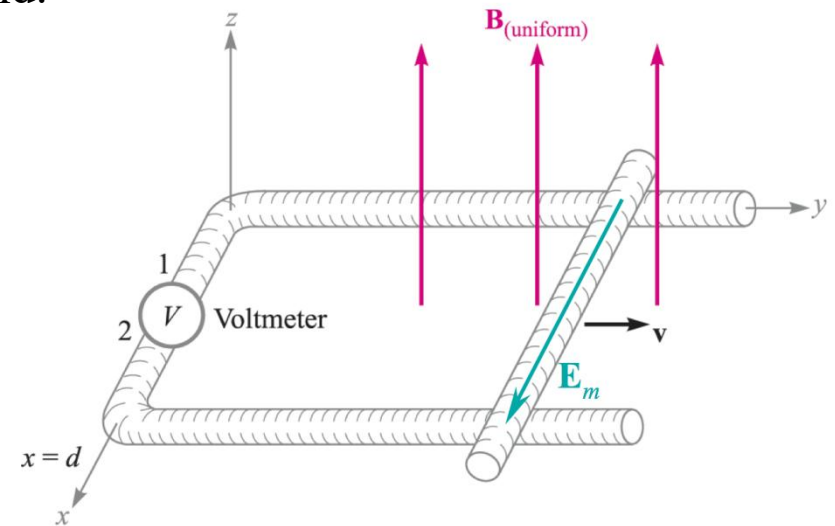
and so.. $\frac{\mathbf{F}}{Q} = \mathbf{v} \times \mathbf{B}$

This result is the *motional field intensity*:

$$\mathbf{E}_m = \mathbf{v} \times \mathbf{B}$$

..and so now the *motional emf* is the closed path integral of $\mathbf{E}_m$ over the same path as before.

$$\text{emf} = \oint \mathbf{E}_m \cdot d\mathbf{L} = \oint (\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{L}$$



Note: the motional field intensity exists only in the part of the circuit that is in motion.

12: Time-Varying Fields

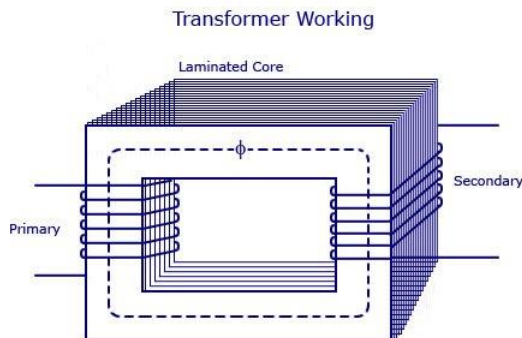
- Two Contributions to emf

For a moving conducting path **in an uniform field**, the emf is composed entirely of **motional emf**:

$$\text{emf} = \oint \mathbf{E} \cdot d\mathbf{L} = \oint \mathbf{E}_m \cdot d\mathbf{L} = \oint (\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{L}$$

When the $\mathbf{B}$ field is time-varying as well, then both effects need to be taken into account.
The **total emf** is:

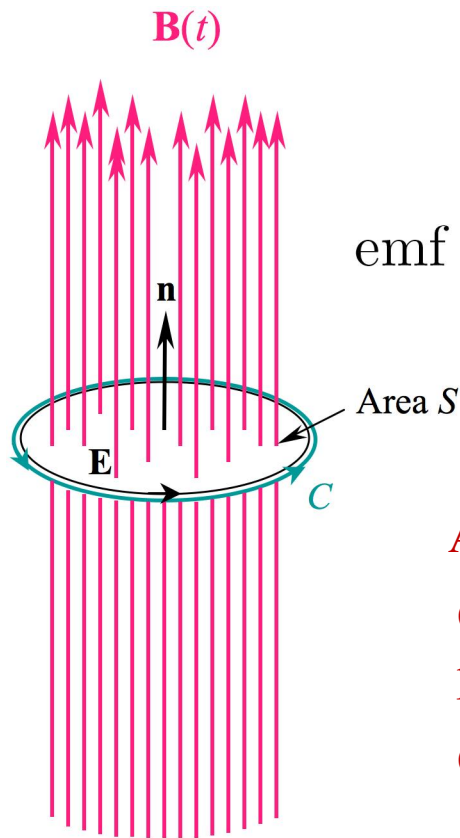
$$\text{emf} = \oint \mathbf{E} \cdot d\mathbf{L} = \underbrace{- \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S}}_{\text{Transformer emf}} + \underbrace{\oint (\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{L}}_{\text{Motional emf}}$$



$$-\frac{d\Phi}{dt}$$

12: Time-Varying Fields

- Summary and rethinking



$$\oint_C \mathbf{E} \cdot d\mathbf{L} = -\frac{d\Phi}{dt}$$

$$\text{emf} = \oint \mathbf{E} \cdot d\mathbf{L} = -\int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S} + \oint (\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{L}$$

A time-changing magnetic field produces an electric field.

Q1: How about a time-changing electric field? Does it produce a magnetic field?

Q2: What is the probable source of magnetic field?



12: Time-Varying Fields

- An Apparent Contradiction

Ampere's Circuital Law: $\nabla \times \mathbf{H} = \mathbf{J}$

Taking the divergence of the second equation gives us a result that is identically zero:

$$\nabla \cdot \nabla \times \mathbf{H} \equiv 0 = \nabla \cdot \mathbf{J}$$

But a previous exercise gave us this result: the equation of continuity:

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t}$$

So what went wrong?

12: Time-Varying Fields

- The Missing Term - Displacement Current Density

Suppose we **add a term** to the right-hand side of Ampere's Law, whose identity we need to determine:

$$\nabla \times \mathbf{H} = \mathbf{J} + \mathbf{G}$$

Added term to be identified

Now take the divergence: $\nabla \cdot \nabla \times \mathbf{H} \equiv 0 = \nabla \cdot \mathbf{J} + \nabla \cdot \mathbf{G}$

From the equation of continuity $\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t}$ This must be true

So now: $\nabla \cdot \mathbf{G} = \frac{\partial \rho_v}{\partial t} = \frac{\partial}{\partial t}(\nabla \cdot \mathbf{D}) = \nabla \cdot \frac{\partial \mathbf{D}}{\partial t}$

..from which:

$$\mathbf{G} = \frac{\partial \mathbf{D}}{\partial t}$$

Now, Ampere's Law is modified:

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

This term is the *displacement current density*

12: Time-Varying Fields

- Symmetry in the Curl Equations

Our new result states that the total current is composed of **conduction current** 传导电流 or **convection current** 运流电流 and **displacement current** 位移电流:

where the **displacement current density** is:

Conduction current density is :

Convection current density is :

In the absence of conduction/convection current, and when **fields are time-varying**, we have:

Note the **symmetry when comparing to Faraday's Law**:

$$\nabla \times \mathbf{H} = \mathbf{J} + \mathbf{J}_d$$

$$\mathbf{J}_d = \frac{\partial \mathbf{D}}{\partial t}$$

$$\mathbf{J} = \sigma \mathbf{E}$$

$$\mathbf{J} = \rho_v \mathbf{v}$$

$$\nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t} \quad (\text{if } \mathbf{J} = 0)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

12: Time-Varying Fields

- Displacement Current in Ampere's Law

Displacement current is found from displacement current density by taking the appropriate surface integral:

$$I_d = \int_S \mathbf{J}_d \cdot d\mathbf{S} = \int_S \frac{\partial \mathbf{D}}{\partial t} \cdot d\mathbf{S}$$

Such an integral of Ampere's Circuital Law curl equation gives:

$$\int_S (\nabla \times \mathbf{H}) \cdot d\mathbf{S} = \int_S \mathbf{J} \cdot d\mathbf{S} + \int_S \frac{\partial \mathbf{D}}{\partial t} \cdot d\mathbf{S}$$

Applying Stokes' Theorem to the left side gives the more familiar form of Ampere's Law, now with displacement current added:

$$\oint \mathbf{H} \cdot d\mathbf{L} = I + I_d = I + \int_S \frac{\partial \mathbf{D}}{\partial t} \cdot d\mathbf{S}$$

12: Time-Varying Fields

• Demonstration of Displacement Current

In this situation, a **time-varying magnetic field** links a wire loop that is connected to a parallel-plate capacitor.

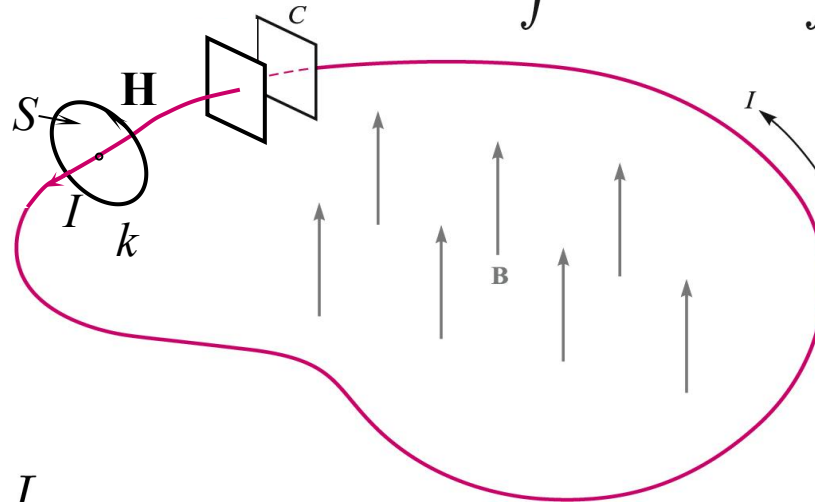
The magnetic field is **presumed time-varying**, and thus it generates emf, which in turn provides the current.

We assume for the moment that **conduction current** I exists in the wire. Ampere's Law is applied to the **circular path** shown, whose **radius is very small compared to the local wire length**. The wire therefore appears very long and straight. We obtain:

$$\oint_k \mathbf{H} \cdot d\mathbf{L} = \int \int_S \mathbf{J} \cdot d\mathbf{S} = I$$

$$\Rightarrow \mathbf{H} = \frac{I}{2\pi\rho} \mathbf{a}_\phi$$

$$\text{emf} = \oint \mathbf{E} \cdot d\mathbf{L} = - \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S}$$



The integration surface is the flat plane whose **perimeter is the circular path, k** . We assign the z direction (locally) to be along the current where it passes through the plane.

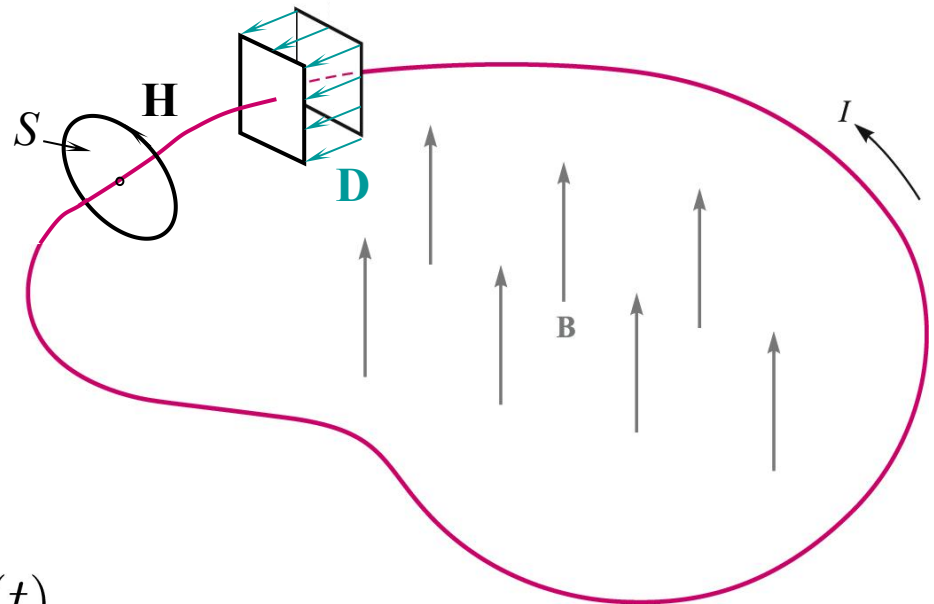
12: Time-Varying Fields

- Demonstration of Displacement Current

While current is flowing, the electric field in the capacitor is changing with time, in step with the plate charge level, deposited by the current.

The capacitor voltage is $V_0(t)$.
Taking the capacitor axis as z ,
assuming free space permittivity,
and with cap dimensions as indicated,
the time-varying electric flux density
within it is:

$$\Phi_e = \int \int \mathbf{D} \cdot d\mathbf{S} = \frac{\epsilon_0 A V_0(t)}{d}$$



12: Time-Varying Fields

• Demonstration of Displacement Current

Now suppose the surface S , surrounded by the original contour, k , is stretched so that it intercepts the electric field between capacitor plates:

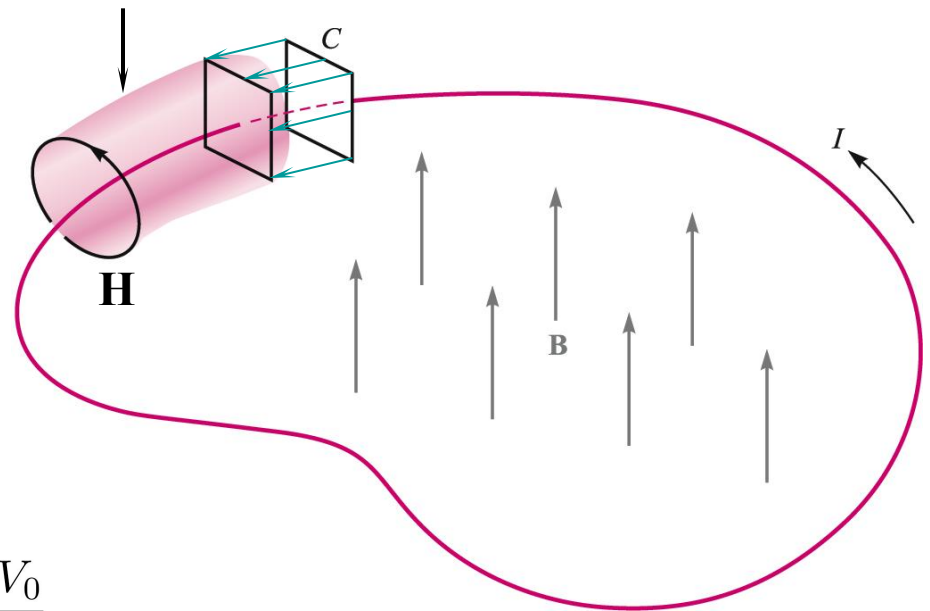
Ampere's Law is again applied, but this time we need to use the displacement current in the capacitor:

$$\oint_k \mathbf{H} \cdot d\mathbf{L} = I_d = \int \int \frac{\partial \mathbf{D}}{\partial t} \cdot d\mathbf{S}$$

$$= \frac{d\Phi_e}{dt} = \left(\frac{\epsilon_0 A}{d} \right) \frac{dV_0}{dt} = C \frac{dV_0}{dt}$$

The result must be independent of the surface that is chosen, which leads to the conclusion that the conduction and displacement currents must be equal:

Stretched surface now intercepts electric field lines, and *no conduction current*



➡ $I = C \frac{dV_0}{dt}$ a familiar result!

13: Maxwell's Equations



- Maxwell's Equations in Point Form

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

Ampere's Circuital Law

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

Faraday's Law of Induction

$$\nabla \cdot \mathbf{D} = \rho_v$$

Gauss' Law for the electric field

$$\nabla \cdot \mathbf{B} = 0$$

Gauss's Law for the magnetic field

13: Maxwell's Equations

- Ampere's Circuital Law in Integral Form

Take the surface integral of the point form equation:

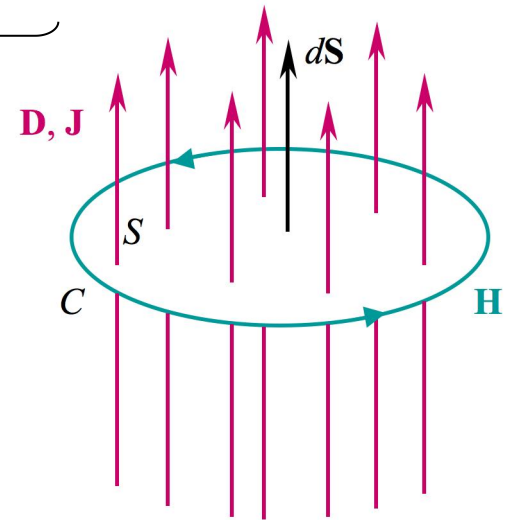
$$\int_S (\nabla \times \mathbf{H}) \cdot d\mathbf{S} = \underbrace{\int_S \mathbf{J} \cdot d\mathbf{S}}_I + \underbrace{\int_S \frac{\partial \mathbf{D}}{\partial t} \cdot d\mathbf{S}}_{I_d}$$

Apply Stokes's Theorem to the left hand side:

$$\underbrace{\oint_C \mathbf{H} \cdot d\mathbf{L}}_{\text{mmf}} = I + I_d$$

where

$$I_d = \int_S \frac{\partial \mathbf{D}}{\partial t} \cdot d\mathbf{S} = \frac{d}{dt} \int_S \mathbf{D} \cdot d\mathbf{S} = \underline{\frac{d\Phi_e}{dt}}$$



Remember to use the right hand convention on relating path direction to the normal vector.

13: Maxwell's Equations



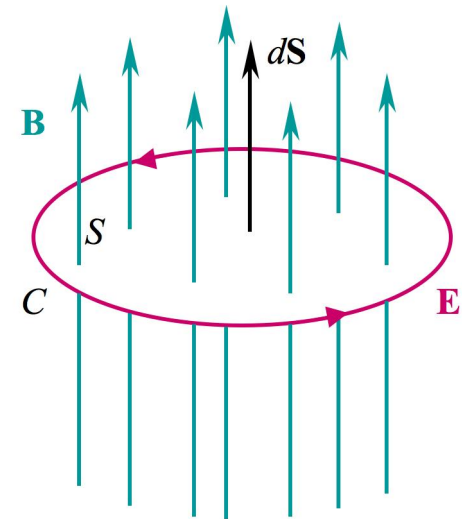
- Faraday's Law of Induction in Integral Form

Take the surface integral of the point form equation:

$$\int_S (\nabla \times \mathbf{E}) \cdot d\mathbf{S} = \int_S -\frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S}$$

Apply Stokes's Theorem to the left hand side:

$$\underbrace{\oint_C \mathbf{E} \cdot d\mathbf{L}}_{\text{emf}} = -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{S} = -\frac{d\Phi_m}{dt}$$



Remember to use the right hand convention on relating path direction to the normal vector.

13: Maxwell's Equations



- Gauss's Law for the Electric Field in Integral Form

Begin with: $\nabla \cdot \mathbf{D} = \rho_v$ and integrate both sides over a volume, v :

$$\int_v \nabla \cdot \mathbf{D} dv = \int_v \rho_v dv = Q_{encl}$$

Apply the divergence theorem to the left hand side:

$$\oint_S \mathbf{D} \cdot d\mathbf{S} = Q_{encl}$$

13: Maxwell's Equations



- Gauss's Law for the Magnetic Field in Integral Form

Begin with: $\nabla \cdot \mathbf{B} = 0$ and integrate both sides over a volume, v :

$$\int_v \nabla \cdot \mathbf{B} dv = 0$$

Apply the divergence theorem to the left hand side:

$$\oint_S \mathbf{B} \cdot d\mathbf{S} = 0$$